

PEDIATRIC ORTHOPEDICS

Developmental Dysplasia of the Hip (DDH)

- Spectrum: **acetabular dysplasia** → **subluxation** → **frank dislocation**.
- **Incidence**: more in Kurds, Sami (Finland); **girls 7:1**; left > right; bilateral 20%.
- **Etiology**: genetic; late-pregnancy hormones (rare in premature); **breech with extended legs**; **tight swaddling/cradleboard** (rare in African carrying with hips abducted).
- **Pathology**: shallow saucer-shaped acetabulum; underdeveloped femoral head ossific centre (slides posterior then up); **anteverted femoral neck**.
- **Neonatal tests**: **Ortolani** (reduces dislocated hip), **Barlow** (dislocates unstable hip).
- **Late features**: clicking hip, limited abduction (difficult nappies), short leg (unilateral), asymmetrical skin creases, late walking, limping/waddling gait.
- **Diagnosis**: clinical; **US (first 3–6 months)**; **X-ray (after 4 months)** — shallow acetabulum, acetabular angle >30°, broken **Shenton's line**, head displaced up/out (above **Hilgenreiner's**, lateral to **Perkins' line**).
- **Management by age**:
- **0–6 mo**: risk factors/+ve test → US; +ve → **Pavlik harness**.
- **6–18 mo**: closed (GA ± traction + adductor tenotomy) or open reduction → **hip spica** (60° flexion, 40° abduction, 20° internal rotation ×3mo).
- **18mo–4yr**: open reduction + remove obstacles ± femoral derotation/shortening ± acetabular osteotomy.
- **4–10yr**: unilateral → surgery up to 10yr (older → AVN risk). **Bilateral after 4yr** → **no operation** (symmetrical, waddling gait tolerated).

Perthes' Disease (Legg-Calvé-Perthes)

- **Idiopathic AVN of proximal femoral epiphysis** in children. 1/10,000; **boys 2:1**; **age 4–8**; bilateral 10%. Assoc: GU malformation, undescended testis, inguinal hernia, Down's.
- **Pathogenesis**: trauma/synovitis → effusion → presses on lateral epiphyseal vessels → AVN. Process takes 2–4 yrs.
- **Clinical**: boy, painful/painless limp, **pain referred to knee**, worse on activity; **+ve Trendelenburg**; loss of abduction + internal rotation.
- **X-ray**: ↑joint space, ↑density ossific nucleus, flattening, fragmentation, lateral displacement, **mushroom head**, **sagging rope sign**.
- **Treatment**: **<6yr** → **non-operative** (activity restriction, physio, abduction brace, crutch). **>6yr** → **containment** (abduction cast/brace ≥1yr, OR varus femoral / innominate osteotomy).

Irritable Hip (Transient Synovitis)

- 1/1000; **boys 2:1**; age 3–9; recurrent 10%. Causes: systemic infection, synovial allergy, trauma.
- **Acute painful hip** (referred thigh/knee/buttock), lasts 5–10 days, ↓internal rotation, healthy-looking ± low fever, **X-ray negative**.
- **Diagnosis of exclusion**: CBC + ESR + CRP + X-ray ± US/MRI. **DDx**: **septic arthritis**, **osteomyelitis**, **Perthes**.
- **Treatment**: bed rest, NSAID, weight-bearing aids.

Femoral Anteversion (In-toe gait)

- Commonest cause of in-toe gait; ↑internal rotation, ↓external rotation. Clumsy gait, trips; **squinting patellae**; sits in "**television position**".
- Improves with growth (anteversion $40^{\circ} \rightarrow 20^{\circ}$ by age 10). Encourage **Buddha position**. Osteotomy rarely needed, not before age 8.

Congenital Talipes Equinovarus (Clubfoot)

- Foot curved down + in: **ankle equinus (plantarflexion), heel varus, forefoot adducted/supinated**. 1–3/1000; **boys 2:1**; bilateral 50%; cause unknown.
- Sole faces posteromedially; small high heel; deep creases; cannot dorsiflex/evert. **Examine for DDH + spina bifida**.
- **Treatment**: start within days of birth. **Ponseti method** (international standard, >90% success): weekly manipulation + casting $\times 5-6 \rightarrow$ percutaneous Achilles tenotomy + cast $\times 3\text{wk} \rightarrow$ foot abduction brace (3mo continuous, then nights $\times 3\text{yr}$). Resistant $\rightarrow$ open release; relapsed older $\rightarrow$ Ilizarov.

PERIPHERAL NERVE INJURIES

Degrees of injury

- **Neurapraxia:** demyelination (mechanical pressure); **full recovery expected**.
- **Axonotmesis:** axon interruption (closed fracture/dislocation); recovery variable.
- **Neurotmesis:** nerve trunk division (open wounds); needs surgical repair.

Clinical & treatment principles

- Features: numbness/tingling, weakness, abnormal posture, **trophic ulcers, loss of sweating**. Electrodiagnostics accurate after few weeks.
- **Closed (neurapraxia/axonotmesis):** wait weeks for spontaneous recovery.
- **Open:** early end-to-end suture or nerve graft. Late presentation → delayed repair.
- Care of paralysed part: full ROM twice daily, protect skin, splint to prevent contracture.
- **Tendon transfer** if no recovery in 18–24 months (donor must have power).

Obstetric brachial plexus injury

- Excessive traction at birth. **Erb's palsy (commonest, C5–6±C7):** paralysis of shoulder abductors/external rotators + supinators → arm **adducted, internally rotated, pronated** ("**waiter's tip**"). **Klumpke's** = lower roots. Physio; surgery if no biceps recovery by 6 months.

Radial nerve injury

- **Low (elbow):** fractures/dislocations → loss of MCP joint extension.
- **High (upper arm):** humerus fracture, tourniquet → **wrist drop** + small sensory loss dorsum thumb base.
- **Very high (axilla):** crutches (crutch palsy), **Saturday night palsy** → triceps paralysis + wrist drop.
- Closed → usually neurapraxia, spontaneous recovery (explore if none by 8–12wk); open → early repair. **Splint wrist in extension while awaiting**. Tendon transfer if no recovery.

Ulnar nerve injury

- **Low (wrist):** laceration/ganglion → hypothenar + interossei wasting, **claw hand**, weak finger abduction, loss thumb adduction (**Froment's test**), sensory loss ulnar 1½ fingers.
- **High (elbow):** fractures, cubitus valgus, **cubital tunnel** → same but **less claw (ulnar paradox)** (long flexors not working).

Median nerve injury

- Supplies most forearm flexors, thenar muscles, lateral 3½ fingers.
- **Low (wrist):** cuts/carpal dislocation → thenar wasting, weak thumb abduction/opposition, sensory loss radial 3½ digits, trophic ulcers.
- **High (forearm/elbow):** + paralysed long flexors thumb/index → **pointing index sign**. Disability mainly from sensory loss.

Lower limb nerves

- **Femoral:** gunshot, traction, thigh bleed → weak knee extension (quadriceps), anterior thigh/shin numbness, depressed knee jerk.
- **Sciatic:** gunshot, stab, hip surgery/dislocation, pelvic fracture → paralysis all muscles below knee, weak knee flexion, sensory loss below knee (except shin), **foot drop**, absent ankle jerk.

- **Common peroneal:** varus stress, splint/POP pressure, leg externally rotated → **foot drop** + sensory loss leg (except sole + shin).
- **Deep peroneal:** anterior compartment syndrome → foot drop + sensory loss 1st web space. **Compartment syndrome** → **immediate decompression**.

High-yield one-liners

- **DDH:** Ortolani reduces, Barlow dislocates; girls 7:1; US <6mo, X-ray >4mo; Pavlik harness 0–6mo.
- **Perthes:** boy 4–8, AVN femoral head, pain referred to knee; <6yr conservative, >6yr containment.
- **Irritable hip** = **diagnosis of exclusion** (rule out septic arthritis).
- **Clubfoot** = **Ponseti method** (>90%); always check for DDH + spina bifida.
- **Neurapraxia** = **full recovery**; **neurotmesis** = **needs repair**.
- **Erb's** = **waiter's tip (C5–6)**. Radial = wrist drop. Ulnar = claw + Froment's. Median = pointing index + thenar wasting.
- **Foot drop** = **common peroneal / sciatic**. Compartment syndrome → urgent fasciotomy.

PEDIATRIC ORTHOPEDICS & PERIPHERAL NERVE INJURIES — Revision Table

Bold = high-yield.

PEDIATRIC HIP & FOOT				
Condition	Key facts	Clinical / signs	Diagnosis	Treatment
DDH	Girls 7:1, L>R, bilateral 20%; breech, tight swaddling	Ortolani (reduces), Barlow (dislocates) ; limited abduction, asymmetric creases, waddling gait	US <6mo, X-ray >4mo ; ↑acetabular angle, broken Shenton's, Hilgenreiner/Perkins lines	0–6mo Pavlik harness ; 6–18mo closed/open reduction + spica; 18mo–4yr open + osteotomy; bilateral >4yr no op
Perthes	Idiopathic AVN femoral head ; boy 4–8, bilateral 10%	Painful/painless limp, pain referred to knee , +Trendelenburg, ↓abduction+IR	X-ray: ↑density, flattening, fragmentation, mushroom head, sagging rope	<6yr conservative; >6yr containment (brace ≥1yr or osteotomy)
Irritable hip (transient synovitis)	Boy 2:1, age 3–9	Acute painful hip, ↓IR, healthy ± low fever, lasts 5–10d	Diagnosis of exclusion (CBC/ESR/CRP/X-ray); DDX septic arthritis	Bed rest, NSAID, weight-bearing aids
Femoral anteversion (in-toe)	Commonest in-toe cause	Squinting patellae, TV position, clumsy gait	Clinical, US, axial CT	Improves with growth; osteotomy rare (not <8yr)
Clubfoot (CTEV)	1–3/1000, boy 2:1, bilateral 50%	Equinus + heel varus + forefoot adduction/supination ; check DDH + spina bifida	Clinical (obvious at birth)	Ponseti (>90%): casts → Achilles tenotomy → abduction brace

PERIPHERAL NERVE INJURY — DEGREES			
Degree	Lesion	Cause	Recovery
Neurapraxia	Demyelination	Mechanical pressure	Full recovery
Axonotmesis	Axon interruption	Closed fracture/dislocation	Variable
Neurotmesis	Nerve division	Open wounds	Needs surgical repair

NERVE LESIONS		
Nerve	Cause	Deficit
Radial	Humerus #, tourniquet, Saturday night/crutch palsy	Wrist drop , loss MCP extension, sensory dorsum thumb base; splint wrist extended
Ulnar	Wrist laceration / cubital tunnel	Claw hand , interossei+hypothenar wasting, Froment's +ve , ulnar 1½ finger loss (high = less claw, ulnar paradox)
Median	Wrist cut / forearm #	Thenar wasting, weak abduction/opposition, radial 3½ finger loss; high = pointing index
Erb's palsy	Birth traction, C5–6±7	"Waiter's tip" (adducted, internally rotated, pronated)
Femoral	Gunshot, thigh bleed, traction	Weak knee extension, anterior thigh/shin numbness, ↓knee jerk
Sciatic	Hip dislocation/surgery, pelvic #	Paralysis below knee, weak knee flexion, foot drop , absent ankle jerk
Common peroneal	Varus stress, splint pressure	Foot drop + sensory loss (except sole + shin)
Deep peroneal	Anterior compartment syndrome	Foot drop + 1st web space loss → urgent fasciotomy

NERVE — TREATMENT PRINCIPLES	
Situation	Action
Closed (neurapraxia/axonotmesis)	Wait weeks for spontaneous recovery
Open	Early suture / nerve graft
No recovery 18–24 mo	Tendon transfer
While awaiting	Full ROM 2×/day, protect skin, splint to prevent contracture